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March 2, 2025

Ratios of consecutive level spacings: Evaluation of a weighted average for infinite-dimensional random matrices from the orthogonal class

I. INTRODUCTION

The average ratio of consecutive level spacings is a common probe for quantum chaos. These notes discuss a simple way to compute a weighted average ratio of consecutive level spacings for infinite-dimensional random matrices from the orthogonal (time-reversal symmetric) symmetry class.

Refs. [1, 2] are usually seen as the pioneering works on the average ratio of consecutive level spacings. A recent study determining the average ratio of consecutive level spacing for infinite-dimensional random matrices for the unitary class by a partially similar approach as the one used here is discussed in Ref. [3].

II. OUTLINE OF THE APPROACH

A. Setting the stage: level spacing distribution for the unitary class

Pedagogical reference: [4, Sec. 2.2]. The Painlevé V differential equation encodes the level spacing distributions for infinite-dimensional random matrices from each of the orthogonal, unitary, and symplectic classes. Written (for later convenience) as a system of two coupled first-order differential equations, it reads

$$\frac{d}{dt}\begin{pmatrix}\sigma\\ \sigma'\end{pmatrix} = \begin{pmatrix}\sigma'\\ -\frac{2}{t}\sqrt{(\sigma - t\sigma')\,(t\sigma' - \sigma + (\sigma')^2)}\end{pmatrix}. \tag{1}$$

In order to obtain the level spacing distributions, (1) needs to be (numerically) solved subject to the boundary condition

$$\lim_{t\to 0^+}\sigma(t) = -\frac{t}{\pi} - \left(\frac{t}{\pi}\right)^2. \tag{2}$$

Setting the boundary at at $t = 0$ has to be avoided because of the division by t in the lower line. This is not a problem, since the boundary condition provides an accurate value of $\sigma(t)$ and $\sigma'(t)$ for $t \to 0^+$.

Fig. 1 shows plots of $\sigma(t)$ and its first two derivatives for illustrational purposes. Because of the kink-like structure of $\sigma''(t)$ around $t \approx \pi$, the numerical algorithm solving (1) is required to work with small step sizes.

The level spacing distribution $P_2(s)$ for unfolded spectra (that is, for spectra with unit density of states) for the unitary class is expressed in terms of $\sigma(t)$ as

$$P_2(s) = \frac{d^2}{ds^2}E_2(s), \tag{3}$$

$$E_2(s) = \exp\left(\int_0^{\pi s}\frac{\sigma(t)}{t}dt\right). \tag{4}$$

Here, $E_2(s)$ gives the probability to have no levels in a randomly selected interval of length s. The subscript "2" indicates the Dyson index β. From the chain rule, it follows that

$$\frac{d^2}{ds^2}e^{f(s)} = \left(\big(f'(s)\big)^2 + f''(s)\right)e^{f(s)}. \tag{5}$$

Taking

$$f(s) = \int_0^{\pi s}\frac{\sigma(t)}{t}dt, \qquad f'(s) = \frac{\sigma(\pi s)}{s}, \qquad f''(s) = \frac{\pi s\sigma'(\pi s) - \sigma(\pi s)}{s^2} \tag{6}$$

gives

$$P_2(s) = \frac{1}{s^2}\left(\pi s\,\sigma'(\pi s) - \sigma(\pi s) + \big(\sigma(\pi s)\big)^2\right)E_2(s). \tag{7}$$

Since solving (1) gives both $\sigma(t)$ and $\sigma'(t)$, the level spacing distribution can be evaluated without the need for additional numerical differentiation.

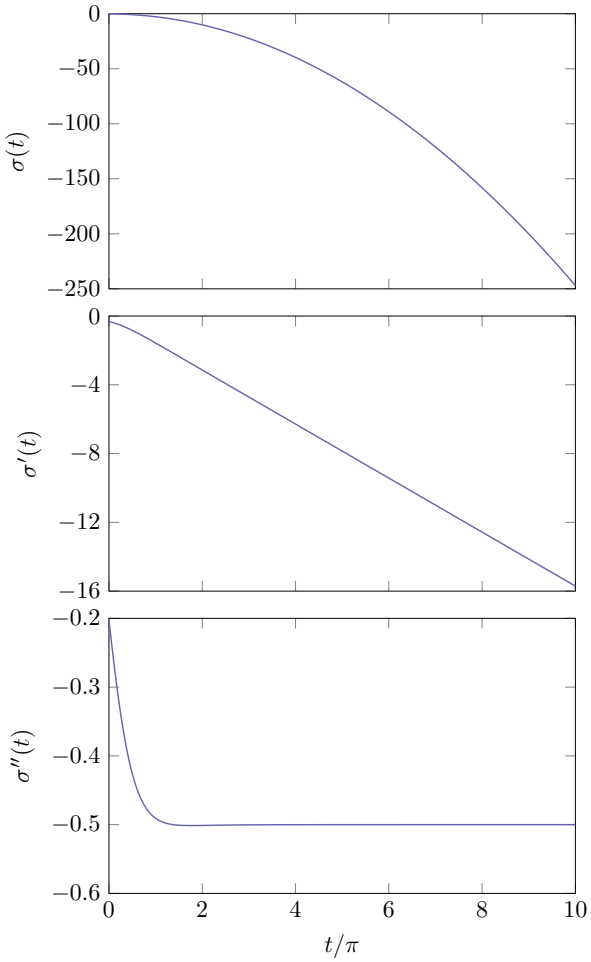

FIG. 1. The numerically obtained solution $\sigma(t)$ of (1) subject to the boundary condition (2), shown together with its first two derivatives $\sigma'(t)$ and $\sigma''(t)$.

B. Level spacing distribution for the orthogonal and symplectic classes

Pedagogical reference: [5]. Eq. (3) is a specific example of the more general expression for the level spacing distribution (after unfolding) of infinite-dimensional random matrices labeled by a Dyson index β, given by

$$P_\beta(s) = \frac{d^2}{ds^2} E_\beta(s). \tag{8}$$

For the orthogonal ($\beta = 1$) and symplectic ($\beta = 4$) classes, one has

$$E_1(s) = \sqrt{E_2(s)}\, \exp\left[-\frac{1}{2} \int_0^{\pi s} \left(-\frac{d}{dt} \frac{\sigma(t)}{t} \right)^{1/2} dt \right], \tag{9}$$

$$E_4(s/2) = \frac{1}{2}\left(E_1(s) + \frac{E_2(s)}{E_1(s)} \right). \tag{10}$$

Analog to (7), expressions for $P_1(s)$ and $P_4(s)$ in terms of $\sigma(t)$ and its derivatives can be obtained through straightforward but somewhat lengthy calculations. One can rewrite (1) as $(t\sigma'')^2 + 4(t\sigma' - \sigma)(t\sigma' - \sigma + (\sigma')^2)) = 0$ to express $\sigma''(t)$ and higher-order derivatives in terms of $\sigma(t)$ and $\sigma'(t)$.

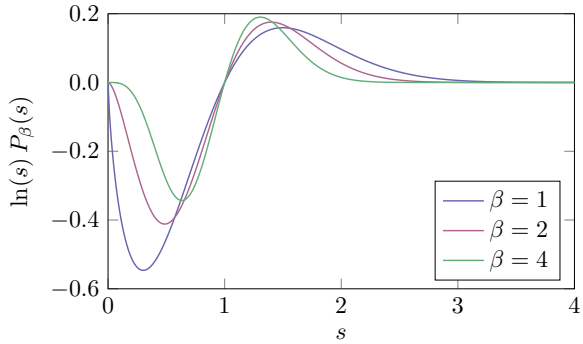

FIG. 2. Plots of $\ln(s)\, P_\beta(s)$ for random matrices of infinite dimension from each of the symmetry classes. All curves are essentially zero for $s \gtrsim 3$, thus allowing for numerical integration over $s \geq 0$.

C. Distribution of two consecutive level spacings for the orthogonal class

Pedagogical reference: [6]. When taking every second level in a spectrum of an infinite dimensional random matrix from the orthogonal class, a spectrum of an infinite dimensional random matrix with mean level spacing 2 from the symplectic class results. Thus,

$$P_1(n = 1, s) = \frac{1}{2} P_4(s/2), \tag{11}$$

where n is the number of ignored levels and $P_\beta(n = 0, s) \equiv P_\beta(s)$. Unrelated to the present discussion, it is mentioned that (11) is only a single example of a larger collection of inter-relations.

D. Average weighted ratio of consecutive level spacings for the orthogonal class

Let $\lambda_1 < \lambda_2 < \lambda_3$ denote three consecutive levels, such that the ratio r of consecutive level spacings is given by $r = (\lambda_2 - \lambda_1)/(\lambda_3 - \lambda_2)$. Here, the focus is on the quantity

$$q = -\ln\left(\frac{\lambda_2 - \lambda_1}{\lambda_3 - \lambda_1}\right) \tag{12}$$

$$= -\ln\left(\frac{r}{1 + r}\right). \tag{13}$$

It is easy to see that the average $\langle q \rangle$ of q for infinite-dimensional random matrices from the orthogonal class can be evaluated as

$$\langle q \rangle = \frac{1}{2} \int_0^\infty \ln(s)\, P_4(s/2)\, ds - \int_0^\infty \ln(s)\, P_1(s)\, ds. \tag{14}$$

Both of these integrals can be evaluated numerically by the approach outlined above. Fig. 2 shows, as an illustration, plots of $\ln(s)\, P_\beta(s)$ for each of the symmetry classes. The point to notice is that all curves go to zero for larger values of s, meaning that the integral can be evaluated straightforwardly.

III. SMALL MATRIX EVALUATIONS

A. Poissonian level statistics

For Poissonian level statistics (uncorrelated levels), the corresponding quantity $\langle q \rangle_{\mathrm{P}}$ can be evaluated analytically. The distribution $P(n = 1, s)$ of two consecutive level spacings is then given by

$$P(n = 1, s) = \int_0^s P(\tilde{s})\, P(s - \tilde{s})\, d\tilde{s} \tag{15}$$

$$= s e^{-s}, \tag{16}$$

where $P(s) = e^{-s}$. This distribution is part of the family of Erlang distributions. Then,

$$\langle q \rangle_{\mathrm{P}} = \int_0^\infty \ln(s)\, s\, e^{-s}\, ds - \int_0^\infty \ln(s)\, e^{-s}\, ds \tag{17}$$

$$= (1-\gamma) + \gamma = 1, \tag{18}$$

as can be evaluated (for example) using `Mathematica`. As pointed out below, this value deviates from $\langle q \rangle \approx 0.81$, thus making $\langle q \rangle$ a valid probe for quantum chaos.

B. Wigner surmise

For Gaussian random matrices of dimension 2, the level spacing distributions are given by the so-called Wigner surmise:

$$P_1(s) = \frac{\pi s}{2} e^{-\pi s^2/4}, \qquad P_4(s) = \frac{2^{18} s^4}{3^6 \pi^3} e^{-64 s^2/(9\pi)}. \tag{19}$$

Then, one can find

$$\langle q \rangle = \frac{4}{3} + \ln\left(\frac{3\pi}{16}\right) \tag{20}$$

$$\approx 0.80409. \tag{21}$$

This value about one percent off from $\langle q \rangle \approx 0.81007$ found for matrices of infinite dimension (see below for details).

C. Wigner-like surmise

For Gaussian random matrices of dimension 3, the distribution $P_1(r)$ of the ratios of consecutive level spacings is given by

$$P_1(r) = \frac{27}{8} \frac{r + r^2}{(1 + r + r^3)^{5/2}}. \tag{22}$$

Ref. [2] contains a derivation. Then, one can find

$$\langle q \rangle = \frac{4}{3} + \ln\left(\frac{3\pi}{16}\right) \tag{23}$$

$$\approx 0.80685. \tag{24}$$

For Poissonian level statistics, one has $P(r) = 1/(1+r)^2$. Then, consistently with the above, one finds

$$\langle q \rangle_{\mathrm{P}} = -\int_0^\infty \ln\left(\frac{r}{1+r}\right) \frac{1}{(1+r)^2}\, dr \tag{25}$$

$$= 1. \tag{26}$$

IV. NUMERICAL EVALUATION

The numerical evaluation consists of two parts. First, some lower moments of the level spacing distributions $P_\beta(s)$ are matched against reference values from the literature. Second, the expectation value (14) of q is determined for random matrices of finite and infinite dimension.

β	N	μ
1	1.00000 00000 00520	1.00000 00000 01404
2	1.00000 00000 00313	1.00000 00000 00037
4	1.00000 00000 04369	1.00000 00000 05754

TABLE I. The numerically obtained normalization N [Eq. (27)] and mean level spacing μ [Eq. (28)] for $\beta = 1, 2, 4$. Values agree up to ~ 12 digits with the expected values, given by unity.

β	σ^2	γ_1	γ_2
1	0.28553 06557	0.68718 99889	0.37123 80638
2	0.17999 38776	0.49706 36204	0.12669 98480
4	0.10409 84222	0.34939 68438	0.02858 27332

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TABLE II. Reference values (top) and numerically obtained values (bottom) for the variance σ^2 [Eq. (29)], skewness γ_1 [Eq. (30)], and kurtosis γ_2 [Eq. (31)] for $\beta = 1, 2, 4$. The reference data in the upper table is taken from Ref. [7], and can be found as well in Ref. [8, Chap. 8]. Values agree up to ~ 9 digits.

A. Lower moments of level spacing distributions

If the procedure outlined above is implemented correctly, the normalization N and mean level spacing μ (the subscript β is omitted for notational convenience) given by

$$N = \int_0^\infty P_\beta(s)\, ds, \tag{27}$$

$$\mu = \int_0^\infty s\, P_\beta(s)\, ds \tag{28}$$

are both equal to unity. Deviations from these values give an indication of the numerical precision of the results. Table I shows the normalization and average of the level spacing distribution for each $\beta = 1, 2, 4$. The results agree with the expected values up to ~ 12 digits.

Ref. [7] discusses numerical evaluations of the variance σ^2, skewness γ_1, and kurtosis γ_2 of the level spacing distributions for infinite-dimensonal random matrices from each of the symmetry classes obtained using a partially complementary method not involving Pinlevé differential equations. These are given by

$$\sigma^2 = \int_0^\infty (s-\mu)^2\, P_\beta(s)\, ds, \tag{29}$$

$$\gamma_1 = \frac{1}{(\sigma^2)^{3/2}} \int_0^\infty (s-\mu)^3\, P_\beta(s)\, ds, \tag{30}$$

$$\gamma_2 = \frac{1}{(\sigma^2)^2} \int_0^\infty P_\beta(s)\, (s-\mu)^4\, ds - 3, \tag{31}$$

respectively. Table II shows this data (top) together with the numerically obtained values (bottom). Values agree typically agree up to all digits.

B. Average weighted ratio of consecutive level spacings and finite-size dependence

The right-hand side of (14) evaluates to 0.8100699328 for random matrices of infinite dimension, which deviates from unity as found for uncorrelated levels, thus making $\langle q \rangle$ a valid probe for quantum chaos. Let $\langle q \rangle_N$ denote

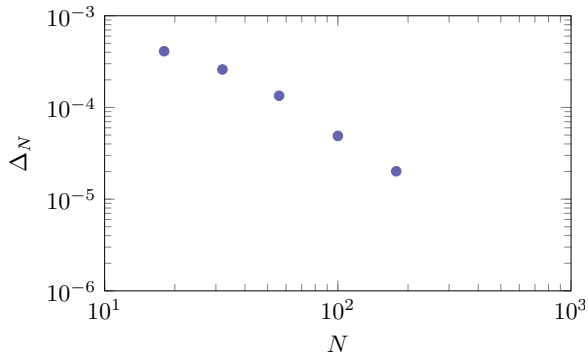

FIG. 3. The deviation Δ_N as defined in (32) as a function of N. Finite-dimensional random matrices have been sampled from the circular orthogonal ensemble.

the analog quantity for random matrices of dimension N. Next, let

$$\Delta_N = \langle q \rangle_N - \langle q \rangle_\infty \tag{32}$$

denote the deviation of $\langle q \rangle_N$ from the asymptotic value.

Fig. 3 shows Δ_N as a function of N. The deviation decreases exponentially with matrix dimension. In the numerical evaluation, finite-dimensional random matrices are sampled from the circular orthogonal ensemble, which has an uniform density of states.

Unrelated to the present notes, it turns out that the function `orthono_group` in `Python` (`scipy.stats` module), which is supposed to generate samples from the circular orthogonal ensemble, actually generates eigenvalue statistics corresponding to the circular unitary ensemble. This is under investigation.

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